

HIV and Oral Care

Integrated Biomedical and Clinical Sciences 2: OMFS 513

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Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Objectives & Competency Statements

1. Understanding the different phases of HIV and the associated clinical manifestations.
2. Discuss the oral manifestations of HIV/AIDS.
3. Review the dental management of patients with HIV/AIDS.

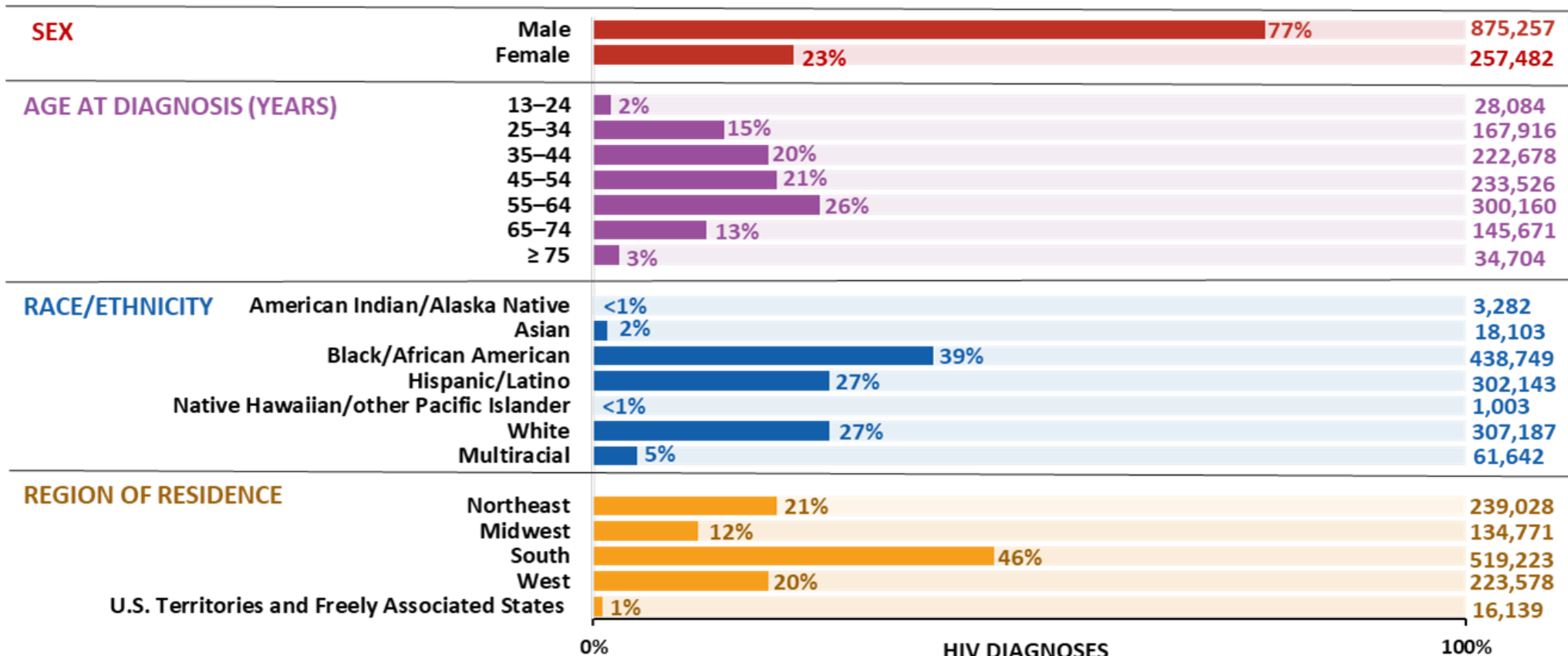
Competency statements met

1.3, 2.1, 3.1, 4.4, and 5.1

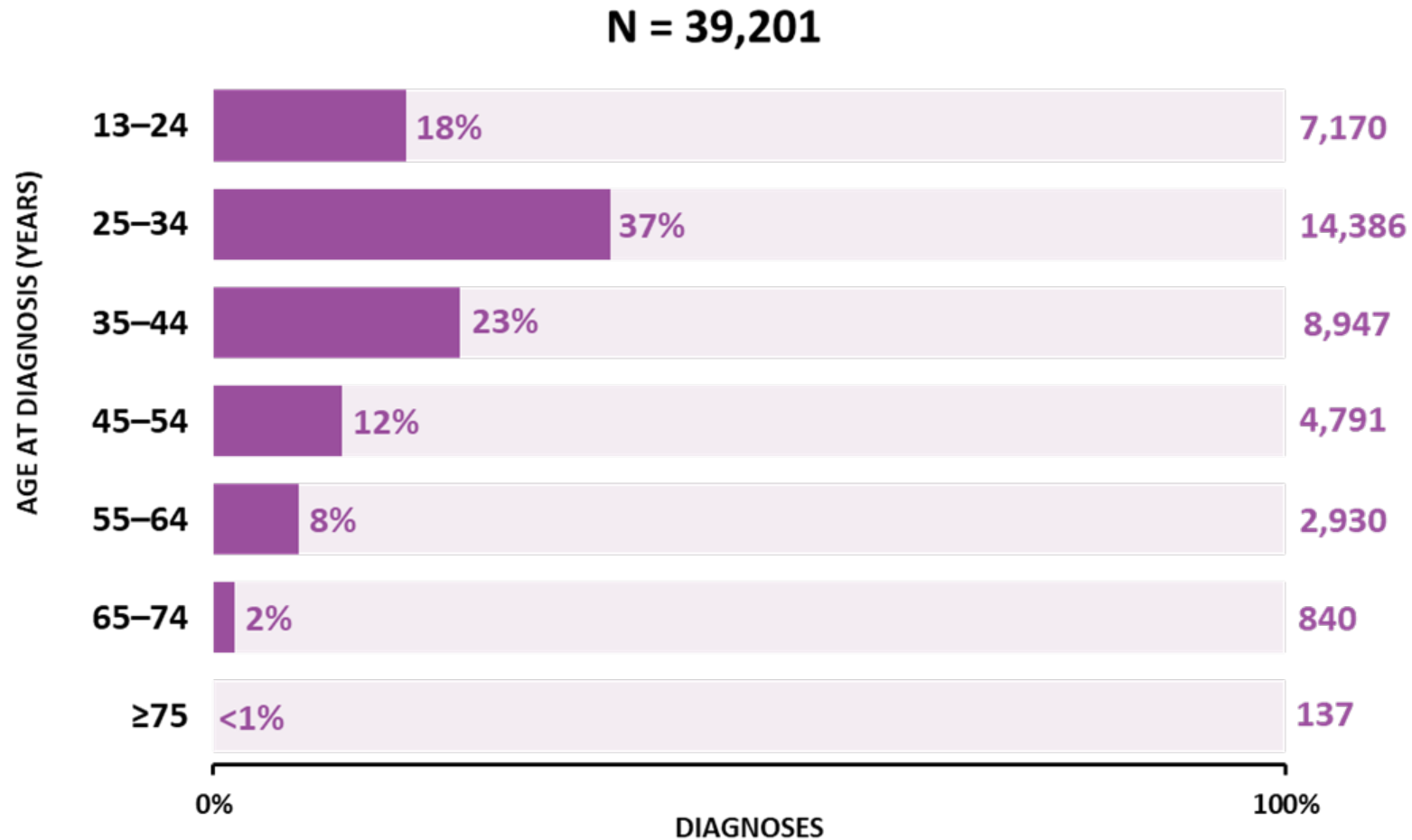
CDC 2023 Statistics on HIV/AIDS

Persons Living with HIV US and Territories

N= 1,132,739



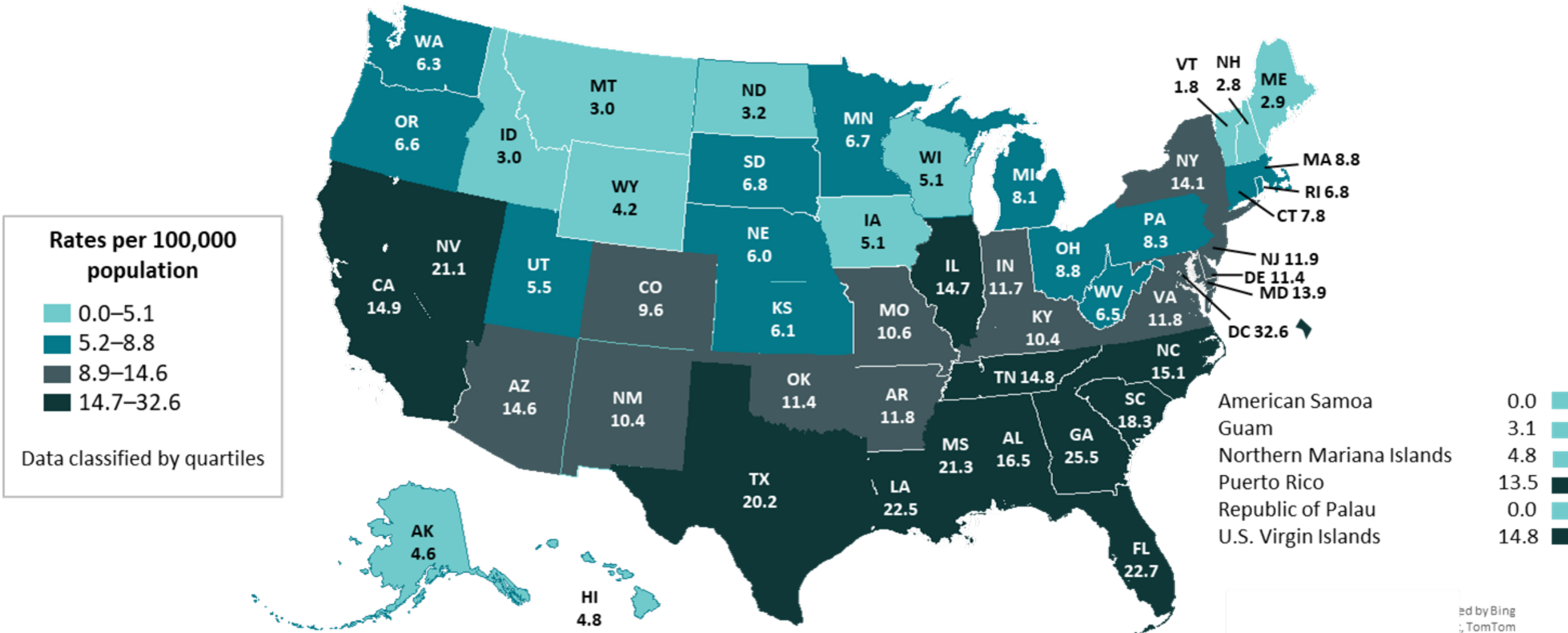
CDC 2023 Diagnosis Statistics on HIV/AIDS - Age



CDC 2023 Diagnosis Statistics on HIV/AIDS

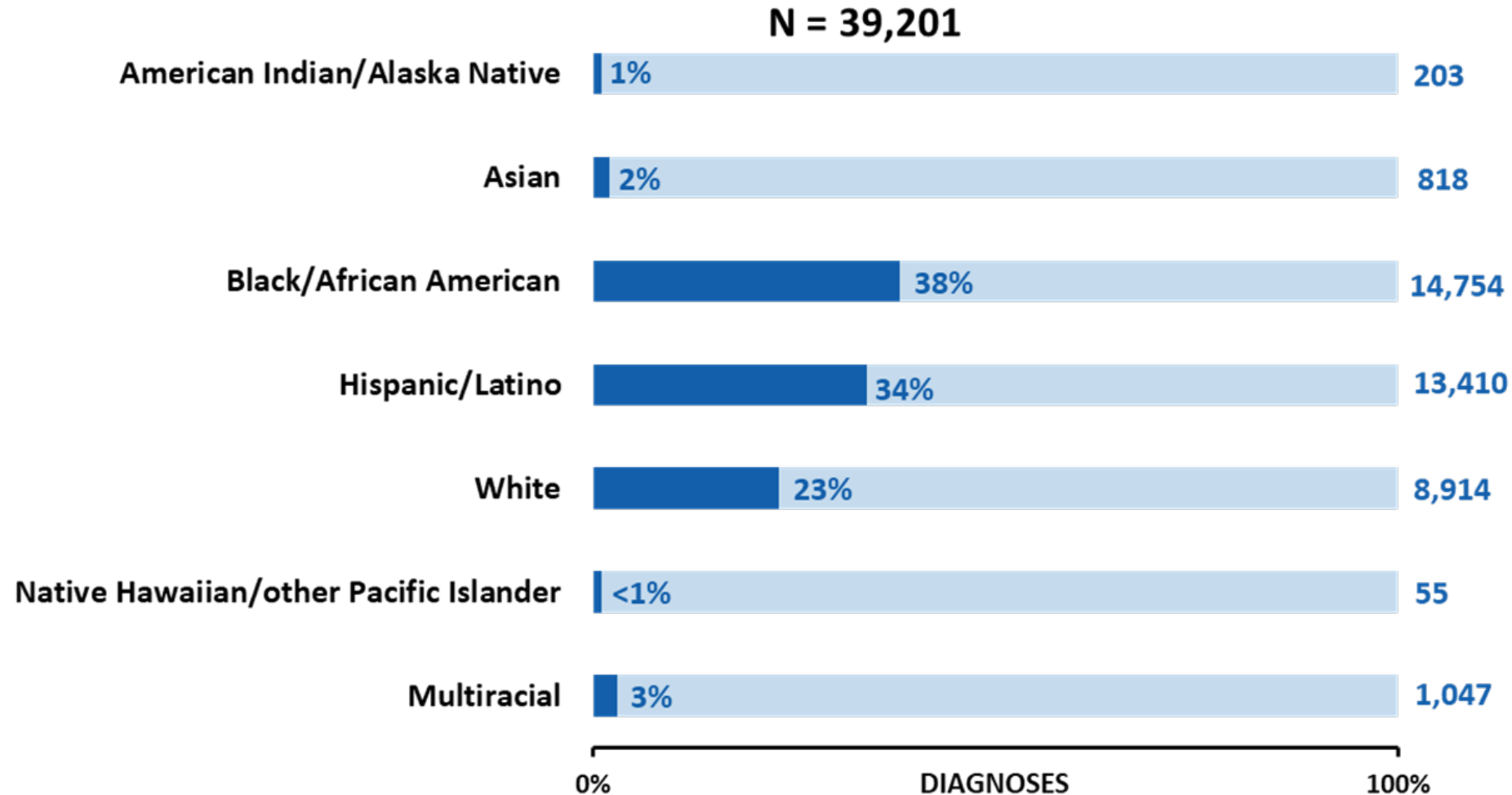
US and US-Territories

N = 39,201
Total Rate = 13.7



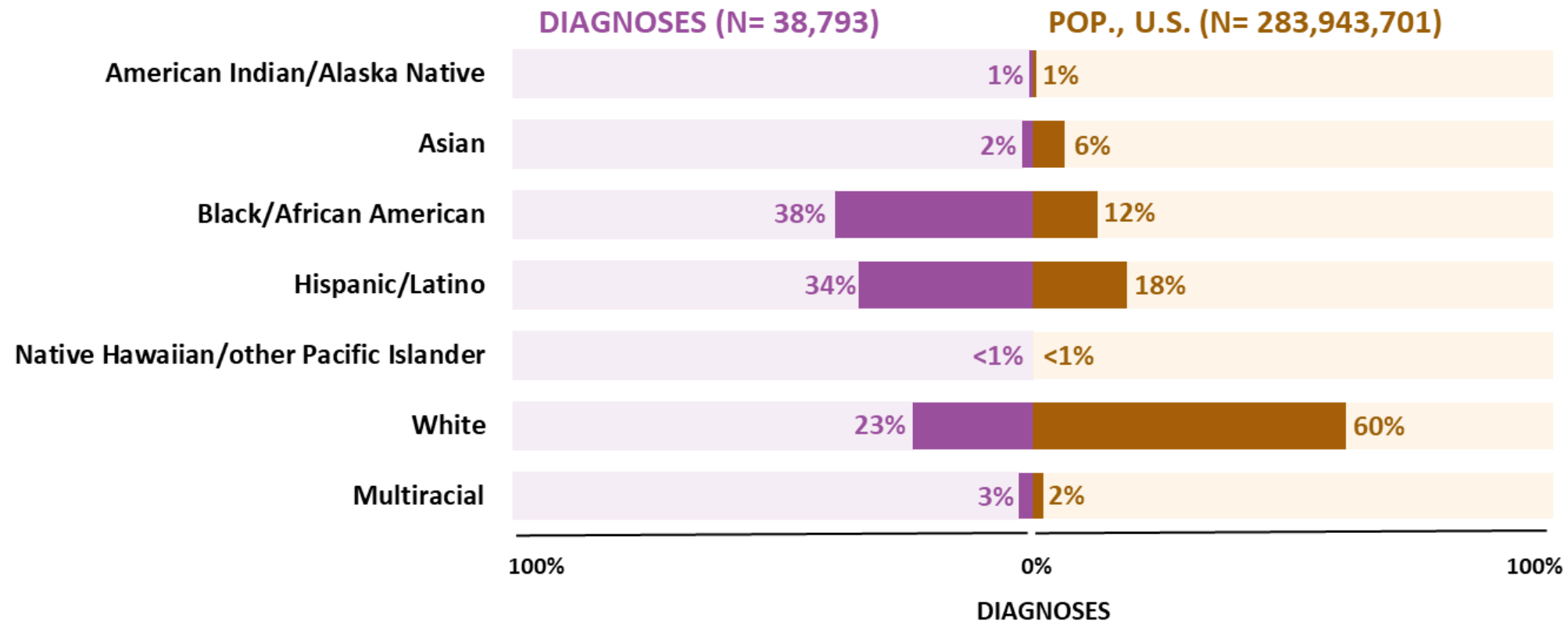
CDC 2023 Diagnosis Statistics on HIV

Race and Ethnicity



CDC 2023 Diagnosis Statistics on HIV

Race and Ethnicity



Phases of HIV/AIDS

- **Acute Phase**
 - Self-limiting illness
 - 3 – 6 weeks post infection
- **Chronic Phase (Latent)**
 - Asymptomatic or mild signs of disease
 - Persistent generalized lymphadenopathy
- **Final (Crisis Phase)**
 - Progressive symptoms of disease
 - CD4 count < 200 cells/microliter= Full blown AIDS
 - Normal CD4+ count=500-1500cells/microliters



Dr. Arthur Ammann, MD-Global Strategies for HIV Prevention [www.hiv.va.gov/provider.image-library/lymphadenopathy]

HIV/AIDS Patient Population

- Patients with unknown HIV status
- Patients with known HIV status & taking ART therapy
- Patients with know HIV status, not taking ART therapy



U.S. Center for Disease Control and Prevention

California Northstate University College of Dental Medicine

Oral Manifestations of HIV/AIDS

- Infectious Diseases
- Aphthous Ulcers
- HIV-Associated Periodontal Disease
- Necrotizing Ulcerative Stomatitis
- Malignant Neoplasms

AIDS-defining illnesses

- • Candidiasis of the esophagus, bronchi, trachea, or lungs [(but NOT the mouth (thrush))
 - Cervical cancer, invasive
- • Coccidioidomycosis, disseminated or extrapulmonary
- • Cryptococcosis, extrapulmonary
 - Cryptosporidiosis, chronic intestinal (greater than one month's duration)
- • Cytomegalovirus disease (other than liver, spleen, or nodes)
- • Cytomegalovirus retinitis (with loss of vision)
 - Encephalopathy, HIV related
- • Herpes simplex: chronic ulcer(s) (more than 1 month in duration); or bronchitis, pneumonitis, or esophagitis
- • Histoplasmosis, disseminated or extrapulmonary
 - Isosporiasis, chronic intestinal (more than 1 month in duration)
- • Kaposi sarcoma
- • Lymphoma, Burkitt's (or equivalent term)
- • Lymphoma, immunoblastic (or equivalent term)
 - Lymphoma, primary, of brain
 - Mycobacterium avium complex or M kansasii, disseminated or extrapulmonary
- • Mycobacterium tuberculosis, any site (pulmonary or extrapulmonary)
 - Mycobacterium, other species or unidentified species, disseminated or extrapulmonary
 - Pneumocystis jiroveci pneumonia
- • Pneumonia, recurrent
 - Progressive multifocal leukoencephalopathy
 - Salmonella septicemia, recurrent
 - Toxoplasmosis of brain
- • Wasting syndrome due to HIV

Infectious Diseases

Candidiasis

- AIDS defining disease
- Oropharyngeal candidiasis is often seen in patients with HIV/AIDS.
 - Extensive and recurrent in clinical presentation
- Types
 - Pseudomembranous “Thrush”
 - Erythematous (Acute & Chronic)
 - Angular cheilitis

Candidiasis



Photo 1

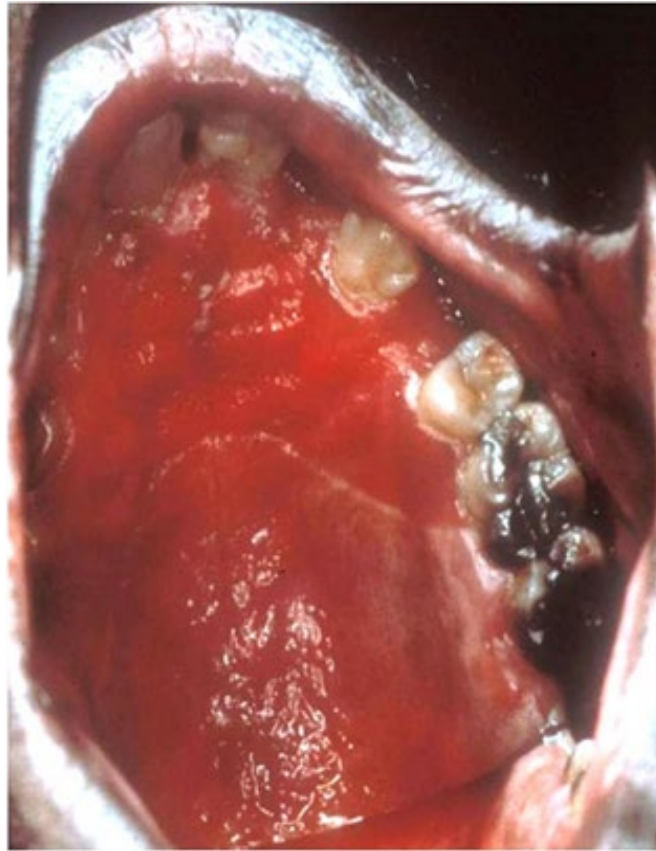


Photo 2

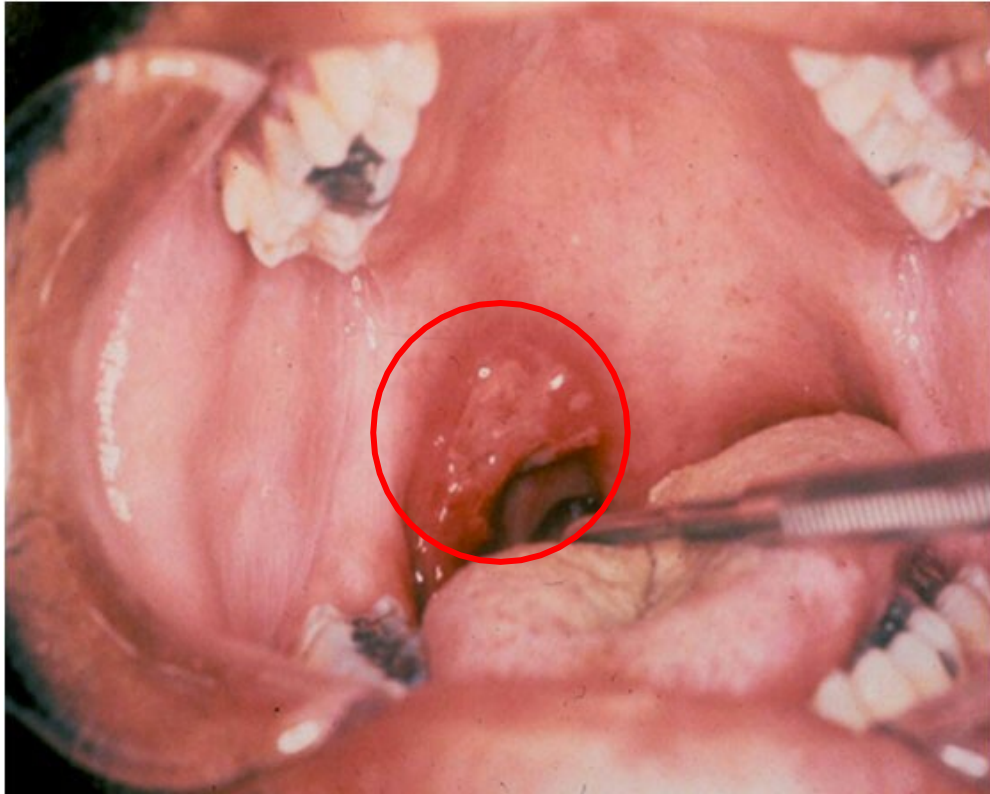


Photo 3

Herpes Simplex

- Herpes simplex is a common infection that can be seen in both immunocompetent and immunocompromised patients. In those who are immunocompromised, the lesions associated with herpes:
 1. Has an atypical pattern
 2. Widespread throughout the oral cavity
 3. Persist for long periods to time
- In immunocompetent patients:
 - Oral herpetic lesions are typically found on keratinized tissues such as the hard palate and the attached gingiva.
 - The lesions are presents for 14 days or less.
- In immunocompromised patients:
 - Oral herpetic lesion can be found on both non-keratinized (buccal mucosa, floor of mouth, or ventral surface of the tongue) and keratinized mucosa.
 - The herpetic ulcers are large and present for weeks or sometimes months before resolving.

Herpes Simplex- Atypical Presentations



Courtesy of Oral Pathology Lab Inc.

Herpes Zoster

- HSV-3 is more infectious in immunocompromised individuals.
- In cases of Zoster (shingles)
 - Involves multiple branches of CNV
 - May exhibit a bilateral presentation

Herpes Zoster



Courtesy of Oral Pathology Lab Inc.

Oral Hairy Leukoplakia



Courtesy of Oral Pathology Lab Inc.

Which virus is associated with oral hairy leukoplakia?

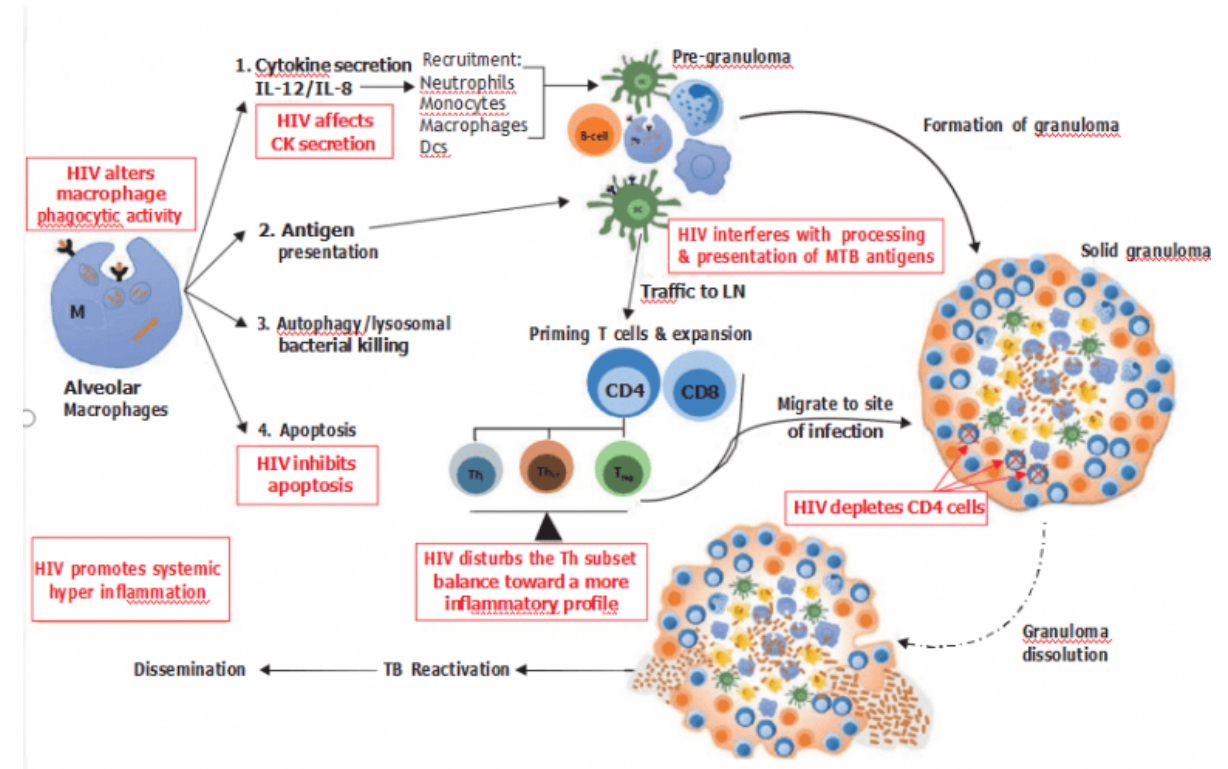
- Epstein-Barr Virus (Human Herpes Virus -4)

Presents as vertical/furrowed white striation on the lateral surface of the tongue

- May extend to involve the ventral and dorsal surface of the tongue.

Co-Infection Concerns

- Tuberculosis
- Hepatitis B and C
- Deep Fungal Infections
 - Cryptococcosis
 - Meningitis
 - Histoplasmosis
 - Coccidioidomycosis
- Persistent infections with a more intense disease state



<https://focustb.com/tuberculosis-and-hiv-co-infection/>

Human Papilloma Virus

- Anogenital warts
- Orofacial papillomas
- High-risk HPV: **16**, 18, 31, 33, 34, 35, 39, 45, 51, 52, 56, 58, 59, 66, 69, and 70

Human Papilloma Virus



Courtesy of Oral Pathology Lab Inc.



Courtesy of Dr. Barbara Griffin

Aphthous Ulcers (Recurrent Aphthous Stomatitis)



Minor Aphthous Ulcer

Females more commonly affected.
Childhood and adolescence.
3-10 mm in diameter.
1-5 lesions
Heals completely without scarring in 7-10 days
Involves labial mucosa, buccal mucosa, ventral surface of the tongue and mucobuccal fold



Major Aphthous Ulcer

Affects male and females equally
Young adults
> 10 mm in diameter
1-10 lesions
Heals completely with scarring in 2-6 weeks
Involves labial mucosa, soft palate, tonsillar fauces.



www.icliniq.com

Herpetiform Ulcer

Female predominance
Seen mostly in adulthood
1-3 mm in diameter
10-100 lesions
Heals in 1-2 weeks
Involves non-keratinized surfaces.

In immunocompetent patients:

1. Found on non-keratinized mucosa
2. Most heal with 2 weeks apart from some major aphthous ulcers.

In immunocompromised patients:

1. Can be located anywhere in the oral cavity
2. Ulcers can last for months and are often large.

Aphthous Ulcer



Courtesy of Oral Pathology Lab Inc.

HIV-Associated Periodontal Diseases

- Linear Gingival Erythema
- Necrotizing Ulcerative Gingivitis
- Necrotizing Ulcerative Periodontitis

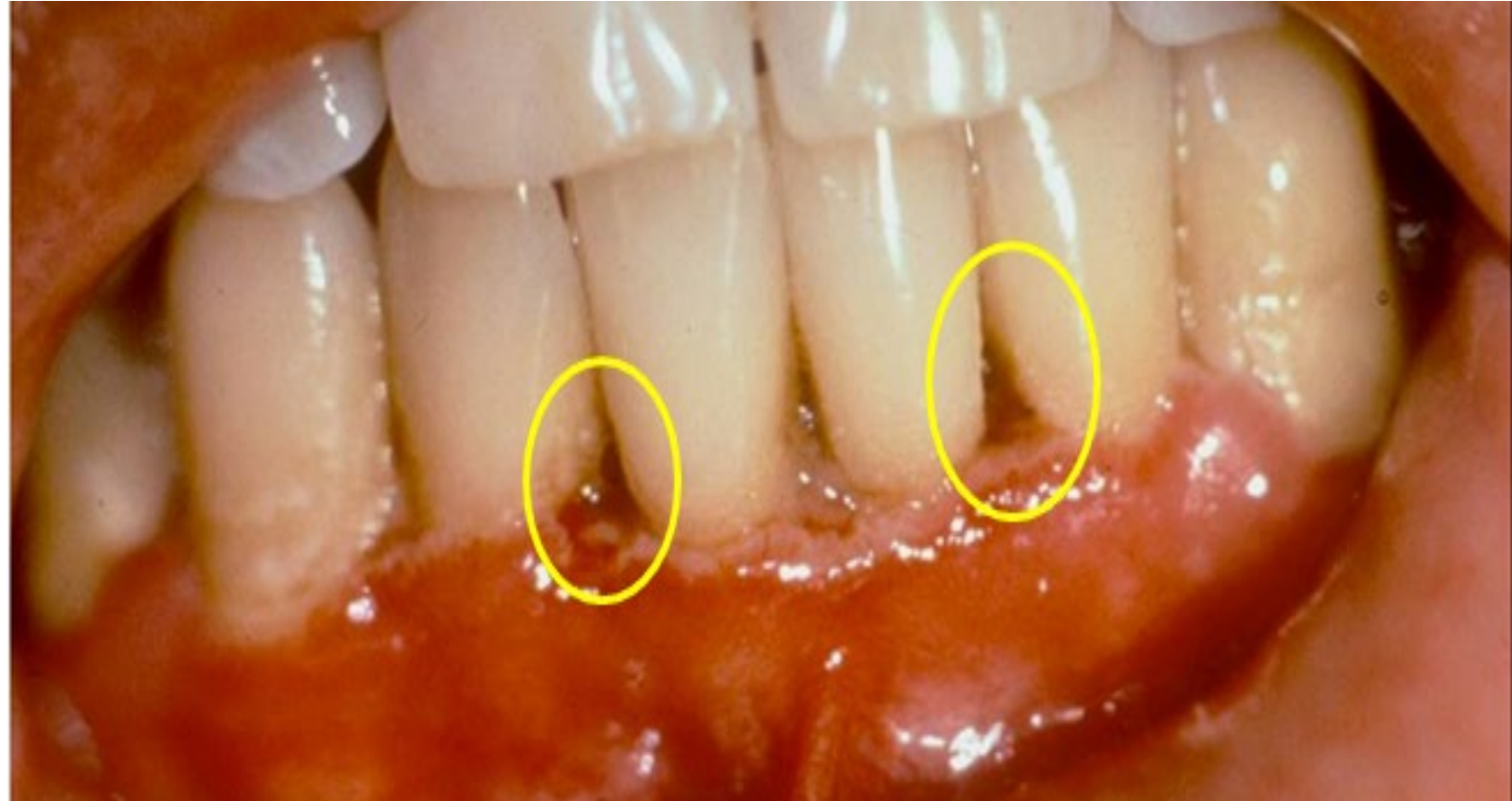
Linear Gingival Erythema



Courtesy of Oral Pathology Lab Inc.

Necrotizing Ulcerative Gingivitis

Necrosis and ulceration of the gingival tissue.
There is no involvement of the underlying periodontal bone



Courtesy of Oral Pathology Lab Inc.

Necrotizing Ulcerative Periodontitis

Necrosis and ulceration of the gingival tissue in addition to the surrounding periodontal ligaments and alveolar bone

- Loss of gingival attachment



Courtesy of Oral Pathology Lab Inc.

Necrotizing Ulcerative Stomatitis



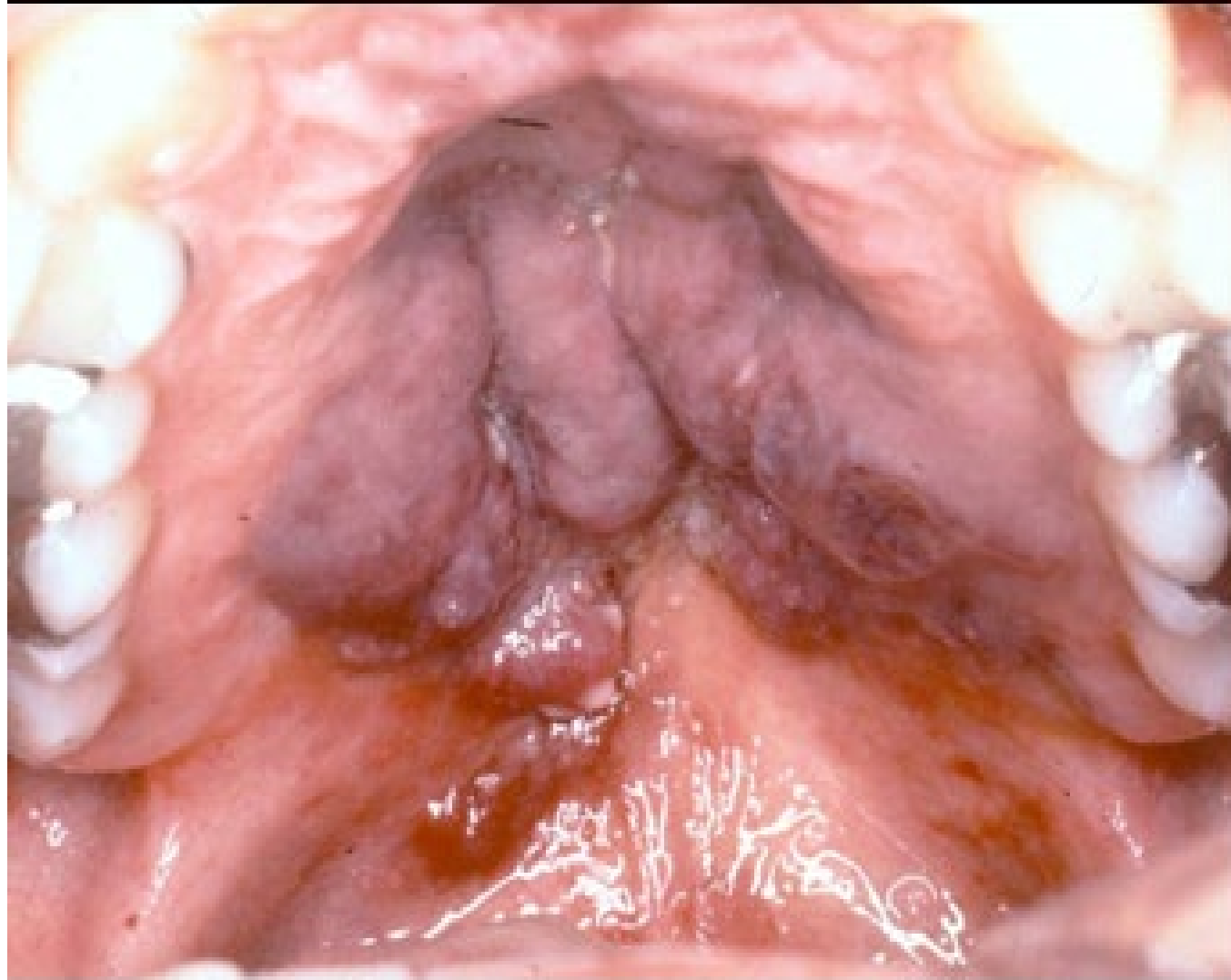
Necrosis and destruction of the mucous membranes in other areas of the oral cavity beyond the mucogingival junction

Neoplasms

Kaposi Sarcoma

- HHV-8
- Male predilection
- Skin: Purple macules or nodules
- Oral: Dark red, brown, or purple macules  masses
- Most common locations:
 - Hard palate
 - Gingiva
 - Tongue

Kaposi Sarcoma



Courtesy of Oral Pathology Lab Inc.

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Lymphoma

- Most common neoplasm in patients with AIDS in the U.S.
- Common HIV-associated lymphomas
 - Non-Hodgkin's Lymphoma
 - Diffuse Large B-Cell Lymphoma
- Can involve the soft tissue and bone of the oral cavity.

Lymphoma



Courtesy of Oral Pathology Lab Inc.

Squamous Cell Carcinoma (SCC)

- HIV+ patients have a higher risk of developing oral cancer when compared to the general population.
 - Conventional and HPV-related squamous cell carcinoma
- The disease is advanced at the time of diagnosis
- Overall guarded/poor prognosis

Conventional and HPV-related SCC



Courtesy of Oral Pathology Lab Inc.



ctvnews.ca/health/hpv

Dental Considerations and Management

- Maintenance of oral health is vital
- Consider the health of the patient when treatment planning
- The appearance of oral lesions may be an indicator of how well the disease is managed.
- Patient education
- Standard precautions and prevention of disease transmission

Dental Considerations and Management

- Standard Precautions
- Postexposure prophylaxis (PEP) after sharps/needle stick incident
- Ask questions about the patient's HIV status?
 1. Does the patient participate in high-risk behavior?
 2. Medications taken and regimen? Are you taking your medications as prescribed by your physician?
 1. Possible adverse reaction to prescribed medications.
 2. Symptoms associated with ART-therapy
 3. What is your current CD4+ count and viral load? Neutrophil count (CBC w/diff)?
 4. Do you have any other medical problems?
 5. Do you have any problems with bruising or prolonged bleeding?

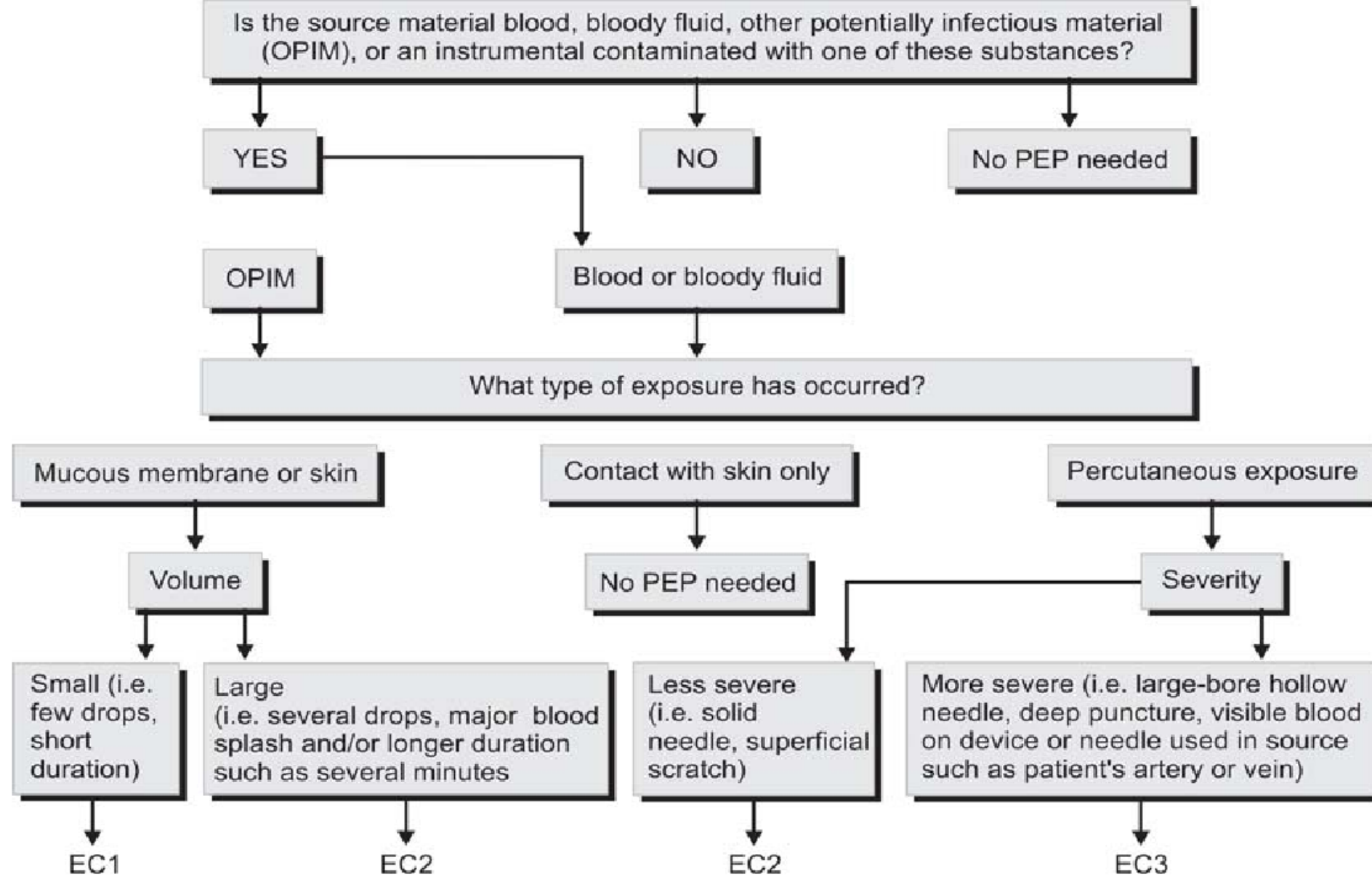


Fig. 1: Determination of exposure code (EC)

National AIDS Control (NACO) guidelines. Available at http://www.nacoonline.org/About_NACO/Policy__Guidelines. Accessed on

Resources

